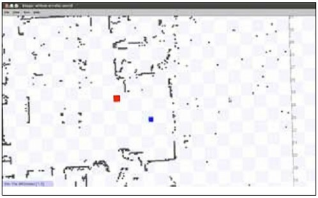

Figure 2: Default stage simulation—the blue square is the erratic robot model

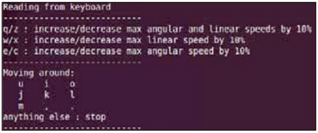

Figure 3: Teleoperation via keyboard

Now, in another terminal window, navigate to the *stage* files using the command given below:

```
roscd stage
```

Note that *roscd* works much like the *cd* command in Bash; this will take you to the location */opt/ros/electric/stacks/simulator_stage/stage*. Navigate to the folder */opt/ros/electric/stacks/simulator_stage/stage/world* and there, start *stage* with a default simulation, as shown below:

```
rosrun stage stageros willow-erratic.world
```

That will result in a pop-up screen as shown in Figure 2; the little blue square is the erratic robot. Now, to start teleoperation, open another terminal window and run the following:

```
rosrun teleop_base teleop_base_keyboard base_controller/
command:=cmd_vel
```

This will give you a set of instructions in the terminal window, as shown in Figure 3. With this terminal window active, try to move the robot around the stage window using the given controls. In another terminal window, *run rxgraph* for a pictorial representation of the *stageros*, *cmd_vel* and other nodes (see Figure 4).

Rviz

Rviz is a 3D visualisation, and it can be used as a ROS node. While the *stage* simulation of the erratic robot discussed in the

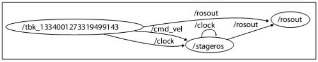

Figure 4: rxgraph showing the nodes

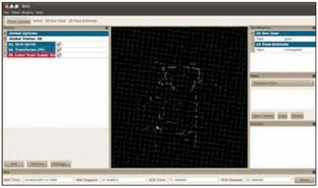

Figure 5: rviz interface

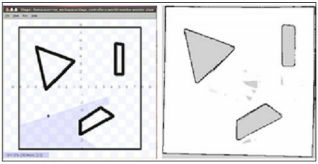

Figure 6: On the right is the actual environment and on the left is the map obtained by SLAM

Figure 7: Stage simulations in rxDeveloper

last section is still running, in another terminal window, run the following:

```
rosrun rviz rviz -d `rospack find stage`/rviz/stage.vcg
```

Rviz will load (Figure 5), and as you move the erratic robot using teleoperation the trajectory can be seen in real-time in *rviz*.

SLAM

Simultaneous Localisation And Mapping (SLAM) is one